

What makes a winner? Predictive features of hockey teams and players

Sports have been around for at least 3000 years (<https://www.thoughtco.com/history-of-sports-1992447>), and they still capture the attention of many people today. A huge part of what continues to draw people in is the practice of prognosticating. Many hours have been spent by fans, discussing past games and attempting to predict future outcomes (<https://playtoday.co/blog/history-of-sports-betting/>). Humans are attuned to finding patterns, and when combined with the competitive interest of sports, discussions can persist endlessly about who the next winner might be. With the use of data science, we can now use actual numbers to back up our guesses.

When we look at National Hockey League (NHL) games, they play a minimum of 82 scheduled games per season (https://www.espn.com/nhl/story/_/id/35265273/sources-nhl-considering-84-game-regular-season-schedule), which gives us plenty of data to mine and analyze. The current system of determining the top team each season is through a playoff system, bringing the determination of “best team” down to one 1v1 game (<https://blog.ticketmaster.com/nhl-playoffs-format-explained/>). This leaves a lot up to chance, and the probability of one bad day for a team or player having huge ramifications. I think there is something interesting to consider with observing teams from a wider view, taking in more than just playoff wins, and considering overall statistics and trends. To place all importance on a playoff game seems to be unfair to a team that may have had a lucky opponent, or a player having a bad day.

The goal of this analysis is to identify overall trends of playoff-winning teams, and to consider defining hockey success in different ways. By observing if players, teams, or specific statistics have a bearing on predicting a playoff winner, recommendations can be made to team managers on strategic moves, or to fans for predictions or betting advice.

The data, in this case, was collected primarily from a Kaggle database (<https://www.kaggle.com/datasets/martinellis/nhl-game-data>), with an additional source of playoff-winning teams per season (<https://www.nhl.com/news/nhl-stanley-cup-champions-winners-complete-list-287705398>). While an overwhelming amount of data was available, the

relevant tables used focused on team stats per game, skater stats per game, game details, and overall team details, including teams that won playoffs.

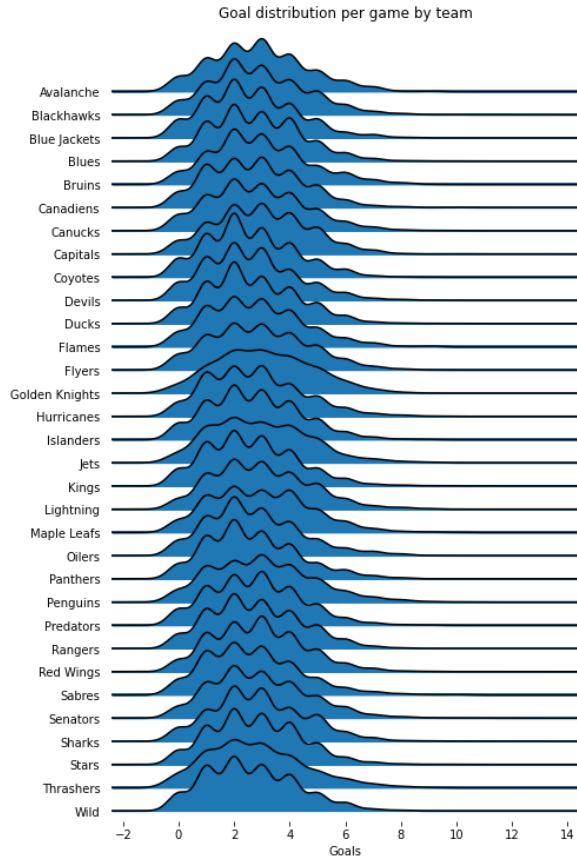
In order to determine the most potent characteristics of a team that is likely to win the playoffs, we are using feature selection through the SelectKBest method. This is a statistical method of feature dimensionality reduction, with the intent of reducing the noise in the dataset to focus on the features with the most correlation with teams that win the playoffs for that season.

After identifying the most influential features, a supervised learning model is created to make predictions on teams and players likely to be a playoff winner. The model is told which teams were winners of their season, and which players were on winning teams for the season. Further refinement of the model provides probabilities of effective predictions.

Features	Score
Todd Reirden	2671.671293
Mike Sullivan	1296.999957
Craig Berube	1291.769381
Larry Robinson	1137.087652
Pat Burns	1076.841306
season	1035.763195
Joel Quenneville	1021.887222
team_id	981.568038
Dave Lewis	706.152694
Darryl Sutter	503.325231
Dan Bylsma	202.991597
Bob Hartley	126.868730
shots	87.290319
Randy Carlyle	75.051682
Barry Trotz	60.740382
John Tortorella	52.625952
Paul Maurice	46.969420
goals	45.795541
Ken Hitchcock	44.982899
Dave Tippett	43.198398

Team features, sorted from most influential to least, of team stats per season that have a predictive strength on determining teams who will win the playoffs. These attributes can be used to make decisions on what makes a successful team.

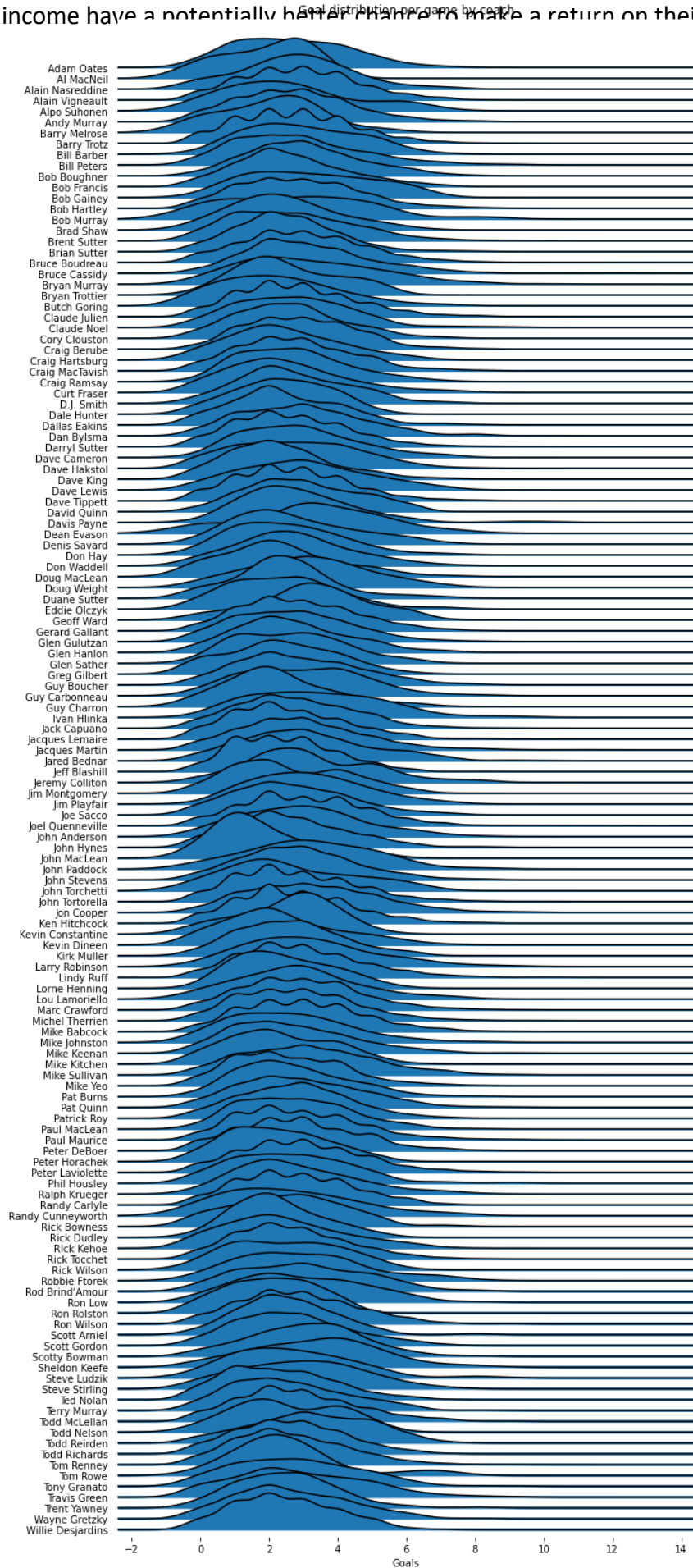
The data gathered from this project can be helpful in two main ways. First, team management and ownership can use the information of which features have the strongest effect when making hiring and recruiting decisions. Knowing that certain coaches tend to lead winning teams may make a strong case for a higher offer of employment. Coaches may last longer in the field than skaters might, but they do eventually age out and are not an ever-present factor. There's no use trying to recruit a coach from 50 years ago for next season's games! So, by further analysis of coaches that tend to lead winning teams, other, new coaches can be identified and hired



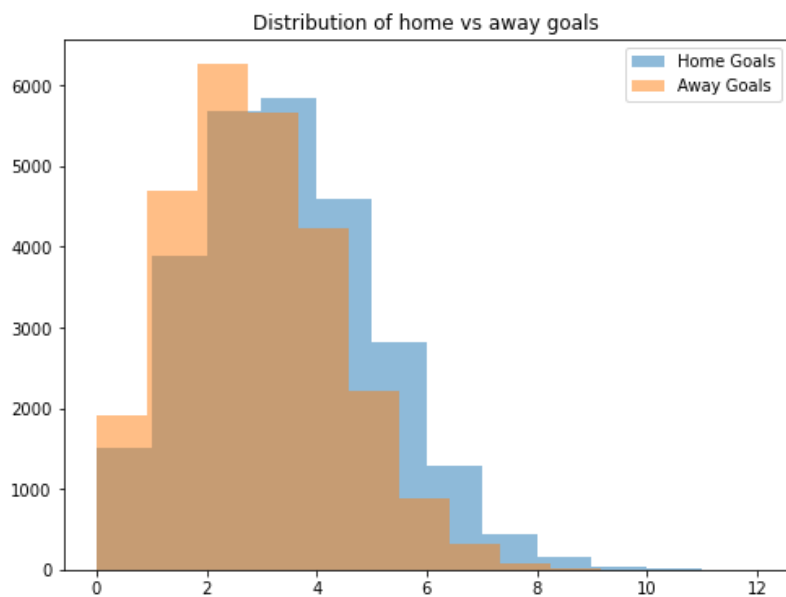
quickly. Other factors and descriptors of skaters can be used to similar effect, identifying potential team members to add to the team, and those to fight to keep on the team.

The other main use of this information is helpful from the sports betting side. With prognostication, betting follows. Odds are an integral part of today's sports betting, and with a refined model, those looking to play the odds for

income have a potentially better chance to make a return on their investment.



In the general analysis of the data, it was expected that there would be some teams and players who showed a significant difference in amount of won games, goals made, or other typical markers of success. Overall, it was seen that there was no significant difference in distribution of goals scored per game when analyzed by team, and no significant difference in the distribution of goals scored under different head coaches. Home team advantage when interpreted through distribution of goals per game also showed a slight, but not very significant difference.



Using feature selection, the most predictive features are primarily coaches. Coach Todd Reirden is the strongest predictor, followed distantly by coach Mike Sullivan. This is interesting, considering that the team Reirden coaches has only won a playoff once in the entire dataset, when the Penguins and Blackhawks have won 3 times each, and their coaches are a lower-scoring predictor.

This model has its limitations, primarily due to the imbalance of classes and relatively short time frame when it comes to seasons. As time goes on, and more data becomes available, the model can be refined further. The main challenge is that these teams and games are dependent on human behavior as much as the people involved. Our model cannot take into account

situations like interpersonal conflicts, or the consideration that a player may not want to leave an area or team for personal reasons. You could collect a team of all of the ideal players, but if they don't work well together, the team is not likely to win the playoffs.

The ethical implications of this project are limited primarily to business decisions on team management, and consumer sports betting. Gambling can be potentially hazardous, especially when using models to make predictions, as some people may place strong faith in the recommendations, and it may encourage reckless gambling activity, when they expect a "sure thing". For these reasons, this model should be used for fun and entertainment, and should expressly state that it is simply a probability.

Potential Audience Questions:

- Will this be able to predict next year's playoff winner?
- Who should we be keeping an eye on for fantasy teams in the next season?
- Why bother creating this if it only has recommendations?
- How does help a team in the recruiting process?
- Why not just look at playoff winners and bet on the recent ones?
- Is there money to be made from this project?
- Is any of this applicable to other sports?
- What can an aspiring professional player use from this project?
- How can there be precautions against excessive sports betting using this model?
- What data are you looking to add in the future to refine the model?

